

1. ✓ A mass of 0.5 kg connected to a light spring of force constant 20 N/m oscillates on a horizontal frictionless surface.

a) Calculate the total energy of the system.

⇒ maximum speed of the mass. if the amplitude of motion is 3 cm.

b) What is the velocity of mass when displacement = 2 cm.

c) Find KE and PE of the system when the displacement = 2 cm.

Solution:-

$$m = 0.5 \text{ kg}$$

$$k = 20 \text{ N/m.}$$

$$x_m = x_0$$

$$i) E = ?$$

$$v_{max} = ?$$

$$x_m = x_0 = 3 \text{ cm} = 3 \times 10^{-2} \text{ m.}$$

$$E = \frac{1}{2} k x_0^2$$

$$= \frac{1}{2} \times 20 \times (3 \times 10^{-2})^2$$

③

$$\Rightarrow V = \sqrt{\frac{k}{m} (x_0^2 - x_1^2)}$$
$$= \sqrt{\frac{20}{0.5} \left\{ (3 \times 10^{-2})^2 - (2 \times 10^{-2})^2 \right\}}$$

$$V = 0.14 \text{ m/sec}$$

i) $x = 2 \text{ cm} = 2 \times 10^{-2} \text{ m}$.

KE = ? when $x = 2 \text{ cm}$

PE = ? when $x = 2 \text{ cm}$.

$$KE = \frac{1}{2} m V^2$$

$$= \frac{1}{2} 0.5 \times (0.14)^2$$

$$= 5 \times 10^{-3} \text{ J}$$

$$PE = \frac{1}{2} k x^2$$

$$= \frac{1}{2} 20 \times (2 \times 10^{-2})^2$$

$$= 4 \times 10^{-3} \text{ J}$$

(2)

$$E = 9 \times 10^{-3} \text{ Joule}$$

$$E = \frac{1}{2} m V_{\max}^2$$

$$\frac{2E}{m} = V_{\max}^2$$

$$\Rightarrow V_{\max} = \sqrt{\frac{2E}{m}}$$
$$= \sqrt{\frac{2 \times 9 \times 10^{-3}}{0.5}}$$

$$V_{\max} = 0.19 \text{ m/sec}$$

ii) $x = 2 \text{ cm} = 2 \times 10^{-2} \text{ m}$

$V = ?$

$$x_0 = x_m$$

$$\text{Total energy} = KE + PE$$
$$\frac{1}{2} k x_0^2 = \frac{1}{2} m V^2 + \frac{1}{2} k x^2$$

$$\frac{1}{2} k x_0^2 - \frac{1}{2} k x^2 = \frac{1}{2} m V^2$$

$$\frac{1}{2} k (x_0^2 - x^2) = \frac{1}{2} m V^2$$

$$\frac{k}{m} (x_0^2 - x^2) = V^2$$

(4)

A force of 5 Newton compresses a spring by 5cm.

- find force constant of spring.
- find the P.E. of compressed spring (0.125J) Ans.

A 100gm object is suspended from a spring whose force constant is 50 N/m

- by how much ~~it~~ does the spring stretch. ($x = 1.9$ Ans)
- What is the period of oscillation of the system. (0.281 sec) Ans
- What is its frequency. ($f = 3.54$ Ans)



$$F = kx$$

$$F = ma$$

$$ma = kx_0$$

0